



Unleash your potential

Module Presenter's Manual

for

**Programming
in C++**

***Effective From: February 2024
Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	February 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Understand Object-oriented Concepts and C++
- Learn the basics of C++
- Learn Flow control statements
- Understand the use of Functions, Pointers and Arrays
- Understand Function Overloading
- Learn about Inheritance
- Learn about Multiple Inheritance and Polymorphism
- Understand data structures using C++ and Exception Handling

2. Information on Session Allocation

Module	Online Hours
C++	32

3. Module Deliverables

To aid the teaching process, following are the deliverables

(Faculties can access eBooks and other components in Onlinevarsity through Aptrack.)

Resources available on Onlinevarsity:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.



Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- A Session is of 2 hours duration.

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 C++ – TL1	Session 2 C++ – TL2	Session 3 C++ – TL3	Session 4 C++ – TL4
2	Session 5 C++ – TL5	Session 6 C++ – TL6	Session 7 C++ – TL7	Session 8 C++ – TL8
3	Session 9 C++ – TL9	Session 10 C++ – TL10	Session 11 C++ – TL11	Session 12 C++ – TL12
4	Session 13 C++ – TL13	Session 14 C++ – TL14	Session 15 C++ – TL15	Session 16 C++ – TL16

C++: Programming in C++

TL: Online Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	C++ - TL1	All the topics as listed below from Session 1 of <i>Object-Oriented Programming With C++</i> Book should be covered in this session. <u>Session 1 - Object Oriented Concepts and C++</u> ➤ Discuss the following: • The Object-Oriented approach • Drawbacks of traditional programming • Object-Oriented Programming ➤ Discuss basic Object-Oriented concepts such as: • Objects • Classes • Properties • Methods • Abstraction • Inheritance • Encapsulation • Polymorphism ➤ Define the structure of a C++ program ➤ Identify the standard input and output functions ➤ Use comments and width(), endl() functions ➤ Describe Private and Public sections of Classes with member functions ➤ Use the Objects and Member functions of a Class	Object-Oriented Programming With C++ SG – Session 1 XP – Session 1 TG – Session 1
2	C++ - TL2	All the topics as listed below from Session 2 of <i>Object-Oriented Programming With C++</i> Book should be covered in this session. <u>Session 2 - Basics of C++</u> ➤ Identify the C++ character set ➤ Discuss the identifiers and keywords ➤ Explain the various data types and qualifiers	Object-Oriented Programming With C++ SG – Session 2 XP – Session 2 TG – Session 2

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Identify the C++ variables ➤ Identify the C++ operators ➤ Understand the precedence and order of evaluation ➤ Discuss mixed mode expressions and implicit type conversions ➤ Understand type casting and type compatibility ➤ Identify C++ shorthand operators	
3	C++ - TL3	All the topics as listed below from Session 3 of <i>Object-Oriented Programming With C++ Book</i> should be covered in this session. <u>Session 3 - Flow control statements</u> ➤ Explain the selection constructs • The if statement • The if...else statement • The if...else if...else statement • Nested if • The switch statement • Nested switch ➤ Identify the iteration constructs • The while loop • The do...while loop • Nested while and do...while loops • The for loop • Multiple initializations/increments in for loop • Nested for loops ➤ Understand simple control statements	Object-Oriented Programming With C++ SG – Session 3 XP – Session 3 TG – Session 3
4	C++ - TL4	All the topics as listed below from Session 4 of <i>Object-Oriented Programming With C++ Book</i> should be covered in this session. <u>Session 4 – Functions, Pointers and Arrays</u> ➤ Identify what functions do and their structure ➤ Discuss the arguments of a function ➤ Discuss return from the function and type of a function ➤ Identify function declaration and function prototype	Object-Oriented Programming With C++ SG – Session 4 XP – Session 4 TG – Session 4

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Understand the sizeof() operator ➤ Discuss call by value and call by reference ➤ Explain recursive functions and identify storage classes ➤ Discuss functions in multifile programs ➤ Express function declaration for extern functions ➤ Discuss and use Pointers ➤ Identify single dimensional arrays ➤ Describe the process of initializing an array ➤ Identify multidimensional arrays	
5	C++ - TL5	The programs taught in Session 1 to Session 4 of <i>Object-Oriented Programming With C++</i> book should be covered in this session.	Object-Oriented Programming With C++ Session 1-4
6	C++ - TL6	All the topics as listed below from Session 5 of <i>Object-Oriented Programming With C++</i> Book should be covered in this session. <u>Session 5 – Function Overloading</u> ➤ Explain the concept of functions with default arguments ➤ Define and use Friend functions with friend classes ➤ Describe function overloading ➤ Describe Operator Overloading ➤ Describe overloading of the Assignment Operator	Object-Oriented Programming With C++ SG – Session 5 XP – Session 5 TG – Session 5
7	C++ - TL7	All the topics as listed below from Session 5 of <i>Object-Oriented Programming With C++</i> Book should be covered in this session. <u>Session 5 – Function Overloading</u> ➤ Describe Copy Constructors ➤ Describe conversion functions which help in conversion from Basic types to User-Defined types from User-Defined types to Basic types between Objects of different Classes ➤ Identify operators that cannot be overloaded	Object-Oriented Programming With C++ SG – Session 5 XP – Session 5 TG – Session 5

Session No.	Session Title	Session Details	Deliverables' Mapping
8	C++ - TL8	All the topics as listed below from Session 6 of <i>Object-Oriented Programming With C++ Book</i> should be covered in this session. <u>Session 6 - Inheritance</u> ➤ Describe single inheritance ➤ Describe Base classes and derived class ➤ Describe how to access base class members and use pointers in classes	Object-Oriented Programming With C++ SG – Session 6 XP – Session 6 TG – Session 6
9	C++ - TL9	All the topics as listed below from Session 6 of <i>Object-Oriented Programming With C++ Book</i> should be covered in this session. <u>Session 6 - Inheritance</u> ➤ Describe types of inheritance ➤ Describe constructors and destructors under inheritance ➤ Describe how to call member functions of the base class and derived class ➤ Describe Container classes	Object-Oriented Programming With C++ SG – Session 6 XP – Session 6 TG – Session 6
10	C++ - TL10	The programs taught in Session 5 and Session 6 of <i>Object-Oriented Programming With C++ book</i> should be covered in this session.	Object-Oriented Programming With C++ Session 5 & 6
11	C++ - TL11	All the topics as listed below from Session 7 of <i>Object-Oriented Programming With C++ Book</i> should be covered in this session. <u>Session 7 - Multiple Inheritance and Polymorphism</u> ➤ Describe multiple inheritance • Constructors under multiple inheritance • Ambiguity in multiple inheritance • Multiple inheritance with a common base ➤ Describe virtual base classes ➤ Use pointers to derived classes	Object-Oriented Programming With C++ SG – Session 7 XP – Session 7 TG – Session 7
12	C++ - TL12	All the topics as listed below from Session 7 of <i>Object-Oriented Programming With</i>	Object-

Session No.	Session Title	Session Details	Deliverables' Mapping
		C++ Book should be covered in this session. <u>Session 7 - Multiple Inheritance and Polymorphism</u> ➤ Describe virtual functions ➤ Describe polymorphism ➤ Describe dynamic binding ➤ Describe pure virtual functions ➤ Describe abstract classes ➤ Describe virtual destructors	Oriented Programming With C++ SG – Session 7 XP – Session 7 TG – Session 7
13	C++ - TL13	All the topics as listed below from Session 8 of <i>Object-Oriented Programming With C++</i> Book should be covered in this session. <u>Session 8 - Data Structures using C++ and Exception Handling</u> ➤ Differentiate between pointers and references ➤ Operate with stacks ➤ Discuss linked lists ➤ Define queues	Object-Oriented Programming With C++ SG – Session 8 XP – Session 8 TG – Session 8
14	C++ - TL14	All the topics as listed below from Session 8 of <i>Object-Oriented Programming With C++</i> Book should be covered in this session. <u>Session 8 - Data Structures using C++ and Exception Handling</u> ➤ Discuss the need of exception handling ➤ Write simple error handling routines ➤ Explain the function terminate() ➤ Explain the function unexpected()	Object-Oriented Programming With C++ SG – Session 8 XP – Session 8 TG – Session 8
15 & 16	C++ - TL15 & TL16	The programs taught in Session 7 and Session 8 of <i>Object-Oriented Programming With C++</i> book should be covered in this session.	Object-Oriented Programming With C++ Session 7 & 8

6. Library References

➤ Practical Programming in C++ by Steve Oualline
➤ The C++ Cookbook by D. Ryan Stephens, Christopher Diggins, Jonathan Turkanis, and Jeff Cogswell
➤ The C++ Programming Language by Bjarne Stroustrup

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